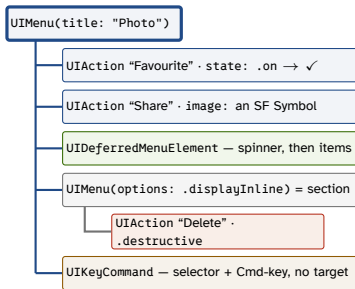
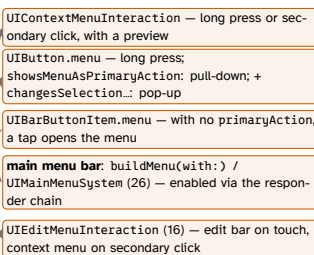


**In one line:** A menu is a **value tree** of UIMenuElements — UIMenu (a group) holding UIAction (a closure), UICommand (a *selector* sent up the **responder chain**) and UIDeferredMenuElement (loaded lazily). The **same tree** is shown by many presenters: a context-menu interaction (with a preview), a button, a bar item, the edit menu, the **main menu bar** (Mac; iPad since iPadOS 26).

## one menu tree (Composite)



## shown by



same UIMenu value everywhere

## context menu, step by step

- 1 long press → configurationForMenuAtLocation (required)
  - 2 highlightPreviewForItemWithIdentifier → a UITargetedPreview lifts the view (visiblePath = its corners)
  - 3 actionProvider(suggestedActions) builds the tree; the previewProvider VC shows · willDisplayMenuFor
  - 4a action tapped → its handler runs, menu closes
  - 4b preview tapped → willPerformPreviewActionForMenuWith — animator.addCompletion { push(detail) }
  - 5 willEndFor + dismissalPreviewForItemWithIdentifier — the identifier finds the cell again
- empty configuration = lift, then cancel ("a menu lives here, nothing now");  
nil = no feedback

## How it works

- **UIAction:** title, subtitle, image, handler; attributes .disabled, .destructive, .hidden, .keepsMenuPresented (16); state .on/.off/.mixed. **UIMenu** options: .displayInline, .singleSelection (radio), .displayAsPalette (17); preferredElementSize .small/.medium/.large (16). Children are fixed: build a new menu and re-assign button.menu.
- **Deferred:** UIDeferredMenuElement { completion in ... } (14) is **cached** after the first load; .uncached (15) re-asks each time. iOS 26 .usingFocus(identifier:shouldCacheItems:): the focused responder's provider(for:) fills it — per-window main-menu items.
- **Context menu:** add a UIContextMenuInteraction(delegate:) to the view, or in a list implement contextMenuConfigurationForItemsAt:point: (16): [] = empty space, several paths = multi-selection. previewProvider nil = snapshot of the view.
- **Main menu:** override buildMenu(with builder:) in the app delegate (13): builder.system == .main, insertChild(\_:atStartOfMenu:), remove(menu:); UIMenuSystem.main.setNeedsRebuild(). iPadOS 26 menu bar: UIMainMenuSystem.shared .setBuildConfiguration(config) { builder in ... }, once at launch; storyboard menu bars unsupported. iOS 18: commands reach iPhone apps too (iPhone Mirroring).
- **Validation = responder chain:** a UICommand has no target — UIKit walks up from the **first responder** to the first one whose canPerformAction(\_:withSender:) is true; adjust title/state in validate(\_ command:). None → disabled.
- **Edit menu:** UIEditMenuInteraction (16) replaces UIMenuController: presentEditMenu(with: UIEditMenuConfiguration(identifier:sourcePoint:)); delegate menuFor:suggestedActions:. Text views: editMenuForTextIn:suggestedActions:.
- **Pointer** (iPadOS 13.4): UIPointerInteraction → UIPointerStyle(effect: .highlight/.lift/.hover); buttons: isPointerInteractionEnabled. **Drag & drop** (11): UIDragInteraction returns [UIDragItem]; UIDropInteraction: canHandle → UIDropProposal(.copy/.move) → performDrop; drag is off by default on iPhone (unverified).

## Example — cell menu with preview, pop-up sort button

```

func collectionView(_ cv: UICollectionView, contextMenuConfigurationForItemsAt
  ips: [IndexPath], point: CGPoint) -> UIContextMenuConfiguration? {
  guard ips.count == 1,
    let id = ds.itemIdentifier(for: ips[0]) else { return nil }
  return UIContextMenuConfiguration(identifier: id as NSString, // NSCopying
    previewProvider: { DetailVC(id: id) } { _ in
      let del = UIAction(title: "Delete", attributes: .destructive) {
        [weak self] _ in self?.delete(id)
      }
      let share = UIDeferredMenuElement.uncached { [weak self] done in
        Task { done(await self?.shareActions(id) ?? []) } // spinner first
      }
      return UIMenu(children: [share,
        UIMenu(options: .displayInline, children: [del])]) } } // a section
    sortBtn.menu = UIMenu(options: .singleSelection, children: sorts.map { s in
      UIAction(title: s.title, state: s == sort ? .off : .on : { _ in pick(s) } })
    })
    sortBtn.showsMenuAsPrimaryAction = true // pull-down ...
    sortBtn.changesSelectionAsPrimaryAction = true // ... + this = pop-up
  }

```

## A button with a menu — what the flags do

| menu | showsMenu... | changes... | behaviour                                      |
|------|--------------|------------|------------------------------------------------|
| nil  | —            | true       | toggle: flips isSelected                       |
| set  | false        | false      | tap = primary action, long press = menu        |
| set  | false        | true       | toggle + long-press menu                       |
| set  | true         | false      | <b>pull-down:</b> menu on touch-down           |
| set  | true         | true       | <b>pop-up:</b> tracks selection, title follows |

## Traps

- ! "The button's menu never shows" — without showsMenuAsPrimaryAction it is a *long-press* menu.
- ! IndexPath captured in a handler or used as the identifier — the diffable snapshot may move it; use the item **id**.
- ! The control retains the menu, the menu its actions: self→button→menu→handler→self is a cycle — [weak self].
- ! Stale items in a deferred element: it is cached after the first load — .uncached.
- ! Menu-bar item always grey: nothing between the first responder and the app answers canPerformAction with true.

## Remember

**"One tree, many presenters; actions carry closures, commands ride the responder chain."** Context menu = configuration → preview → menu → commit.